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LOGIC_MIXED_MODE11-1P8M-MMC_ AL_L110AE-CALIBRE-LVS-1.2_P4

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Revision History

Revision	Date	Editor	Description
1.2-P4	05/03/2019	Wendy Tsai	1.Add variables of "POWER_NAME" and "GROUND_NAME" for global power and ground name setting 2.Enhance DRC_1 boolean. 3.Enhance ERC check "WRONG_FLOATING_PSUB". 4.Add "WIDE_DEVICE_TERM_RESISTANCE: RES" in CCI_star_cmd. 5.Add "CALIBRE_LVS_DEVICE_TYPE_MOS" in CCI_star_cmd. 6.Add "genericMos" comments in techgen file. 7. Amend PEX ALIAS and enhance boolean of ME1_ALL to ME8_ALL. 8.Modify MOSCAP pin
1.2-P3	10/20/2017	Lise Lin	1. Follow G-06 reversion 2. Add device: n_50ud18_lvtl110ae, p_50ud18_lvtl110ae 3. Amend boolean of: N_LDCD_D50G33_L110AE/P_LDCD_D50G33_L110AE N_LDCD_D85G33_L110AE/P_LDCD_D85G33_L110AE 4. Amend MIMCAP boolean
1.2-P2	06/12/2017	Lise Lin	1. Add 5V Native NMOS(N_NVT_50_L110AE) inside 5V_RVT IO voltage option. 2. Follow G-05 MC Model to Add NETLIST ELEMENT "X" to devices in PEXRUN_MC_HSPICE. 3. Add #SEPARATE_GND switch (default close) Please refer to Note 14 for the usage. 4. ERC enhancement. 5. Modify Seed Layers boolean. 6. QCI mapping file updated for sheet resistance.
1.2-P1	11/09/2016	Chia Cheng Chen	1. Add 5V Native NMOS(N_NVT_50_L110AE) inside 5V_LVT IO voltage option. 2. Modify cf of MIMCAPS_10F_MML110E, MIMCAPS_15F_MML110E and MIMCAPS_20F_MML110E. 3. Follow G-05 to Add NETLIST ELEMENT "X" to N_25_L110AE, P_25_L110AE and N_BPW_25_L110AE. 4. Follow G-06 to Add VTN_NC(14:36).
1.1-P4	08/18/2016	Chia Cheng Chen	1. Modify the techgen_scr for blocking capacitance between top and bottom plate of MIMCAP. 2. Enhance the boolean of NTAP, PTAP and TTAP.
1.1-P3	11/30/2015	Chia Cheng Chen	1. Modify the boolean of metal to make syntax check pass. 2. Add TOOL_VERSION_AFTER_CALIBRE_2010 switch.
1.1-P2	10/01/2015	Chia Cheng Chen	1. Enhance "PEX IGNORE" for VIA of MOMCAP. 2. Add BLOCKING layer in QRC flow to enhance accuracy for NCAP/PCAP/RF device. 3. Modify CCI mapping file, QCI mapping file and techgen file for 1P6M1T and M56_MMC options. 4. To extract the capacitance of BM-to-body of Base-Band MOMCAPS, separate MEX_I into MEX_I and MEX_I_MOM. 5. Base on G-05 to modify rs_R_2X_L110E from 40e-3 to 35e-3. 6. Follow G-05 revision to add following devices : -NPN_SV45X45_L110AE -NPN_SV90X90_L110AE -NPN_SV135X135_L110AE
1.1_P1	02/05/2015	Eric Chang	1.Follow G-05 to designate the model name of M1,1X,2X metal resistors as

			R_M1_L110E, R_1X_L110E, R_2X_L110E. 2.Add note2 and note3. 3.Follow G-06 to allow RNNPO_L110E, RNPPPO_L110E on T_WELL.
1.0_P3	08/21/2014	Eric Chang	1.Modify AREA and PJ to W and L for the following diodes extracted in layout netlist to support extracted view in QCI flow. DIOP_25_L110AE, DION_25_L110AE, DIOP_33_L110E, DION_33_L110E, DIOP_11_L110E, DION_11_L110E, 2.Follow G05 to modify MIMCAPS_2F_MM's capsw to 0.0374e-9.
1.0_P2	10/04/2013	Deborah Huang	1.Add switch "RCFLOW_QCI" to support Calibre+QRC flow 2.Add Inductors as below L_CR40K_RFVIL_1P6M2T40K L_SQSK_RFVIL_1P6M2T40K L_SY40K_RFVIL_1P6M2T40K L_SY40KCT_RFVIL_1P6M2T40K
1.0_P1	08/22/2013	Deborah Huang	1.Add switch "CHECK_DIFF_Used_As_Conductor",refer to note1 2.Add 110nm_CCI_star_cmd/query.txt for CCI flow 3.Added N_LDCD_D50G33_L110AE/P_LDCD_D50G33_L110AE N_LDCD_D85G33_L110AE/P_LDCD_D85G33_L110AE(add layer SIZE17(118;34)) 4.Added LVS BUILTIN DEVICE PIN SWAP NO 5.Added MOSPROP_LD for LDMOS 6.Modify MMC connection for CCI flow
0.2_P7	03/26/2013	Josh Hong	1.Added new MMC option(default:NONE_MMC) 2.Modifed MIMCAP_20F Cf from "3.74E-11" to "4.95e-11" 3.Fixed N_PD141_HSSPL110E L from "0.13u" to "0.133u"
0.2_P6	11/15/2012	Josh Hong	1.Two or more RF MIMCAPs can't use the identical well 2.Allow RF MIMCAP to be located between 1X & 2X metal 3.Add 5V IO MOS as follows: N_50_LVTL110AE P_50_LVTL110AE N_50_RVTL110AE P_50_RVTL110AE 4.Add two switches: 5V_LVT 5V_RVT
0.2_P5	08/16/2012	Josh Hong	1.Modified "PWELL" to "PSUB_ALL" 2.Modified RF MIMCAP boolean
0.2_P4	08/03/2012	Josh Hong	1.Follow G-05 revision to add following devices: DION_25_L110AE DIOP_25_L110AE NCAP_25OD33 N_BPW_25OD33_L110AE N_BPW_25UD12_L110AE N_BPW_25UD18_L110AE N_BPW_25_L110AE PCAP_25OD33 L_CR8K_RFVIL_1P5M1T L_SQSK_RFVIL_1P5M1T L_SY8KCT_RFVIL_1P5M1T L_SY8K_RFVIL_1P5M1T N_33_HGRF P_33_HGRF

			N_33BPW_HGRF N_NVT_25_L110AE N_NVT_25OD33_L110AE 2.Allow RF MOMCAP in all metal option 3.Modified NMK layer to allow MOS with SAB in IOID
0.2_P3	02/06/2012	Josh Hong	1. Base on G-05 to add the following devices: . 3.3V HG IO NMOS(N_33_HGL110E) . 3.3V HG IO PMOS(P_33_HGL110E)
0.2_P2	01/20/2012	Josh Hong	1. Base on G-05 to add the following devices: HSSP 1.2V 6T SRAM 1.41um^2 (maker layer 98(2)) . Pull Down MOS (N_PD141_HSSPL110E) . Pass Gate (N_PG141_HSSPL110E) . Load PMOS (P_L141_HSSPL110E) LL 1.2V 6T SRAM 1.41um^2 (maker layer 98(4)) . Pull Down MOS (N_PD141_LLL110AE) . Pass Gate (N_PG141_LLL110AE) . Load PMOS (P_L141_LLL110AE)
0.2_P1_01	12/30/2011	Josh hong	1. Base on G-05 to add the following devices: N_25UD12_L110AE (Marker layer: 85,26) P_25UD12_L110AE (Marker layer: 85,26) N_25UD18_L110AE (Marker layer: 85,25) P_25UD18_L110AE (Marker layer: 85,25)
0.1_P3_03	08/29/2011	Ansel Huang	1. Base on G-05 to add the following devices: L_CR8K_RFVIL_1P6M0T1F L_SQSK_RFVIL_1P6M0T1F L_SY8K_RFVIL_1P6M0T1F L_SY8KCT_RFVIL_1P6M0T1F LNW_CR8K_RFVIL_1P6M0T1F LNW_SQSK_RFVIL_1P6M0T1F LNW_SY8K_RFVIL_1P6M0T1F LNW_SY8KCT_RFVIL_1P6M0T1F L_CR40K_RFVIL_1P6M0T1U L_SQSK_RFVIL_1P6M0T1U L_SY40K_RFVIL_1P6M0T1U L_SY40KCT_RFVIL_1P6M0T1U LNW_CR40K_RFVIL_1P6M0T1U LNW_SQSK_RFVIL_1P6M0T1U LNW_SY40K_RFVIL_1P6M0T1U LNW_SY40KCT_RFVIL_1P6M0T1U L_CR20K_RFVIL_1P7M0T1F1U L_SQSK_RFVIL_1P7M0T1F1U L_SY20K_RFVIL_1P7M0T1F1U L_SY20KCT_RFVIL_1P7M0T1F1U L_CR12K_RFVIL_1P8M0T1F1U L_SQSK_RFVIL_1P8M0T1F1U L_SY12K_RFVIL_1P8M0T1F1U L_SY12KCT_RFVIL_1P8M0T1F1U N_25_L110AE N_25OD33_L110AE P_25_L110AE P_25OD33_L110AE N_12_LLRF N_12_SPRF N_12BPW_HSRF N_12BPW_LLRF N_12BPW_SPRF

			N_33BPW_RF P_12_LLRf P_12_SPRF MIMCAPS_2F_PSUB_RFKF MIMCAPS_2F_NWELL_RFKF 2. Add 2.5V_IO switch 3. Base on G-05 to update mimcap name from MIMCAPS_20F_MM to MIMCAPS_2F_MM 4. Base on G-05 to add PEXRUN_MC_HSPICE option for DIODE and PCAP 5. Base on G-05 to update MOMCAPS_ARRAY_VP5_RFVCL from 4 terminals to 3 terminals 6. Base on G-06 to update boolean operation of Transformer 7. Modify source terminal of RF MOS from NSD_I/PSD_I to NSD_I_S/PSD_I_S 8. Update CCI mapping file 9. Update PEX IGNORE command
0.1_P3_02	06/10/2011	Jacob Huang	1. Add Back-To-Back diode for optional usage. 2. Modified the PCAP_33.
0.1_p3_01	02/14/2011	Jacob Huang	1. Add DIOPW and DIODNW due to SPICE update. 2. Add PNP_SV90X90_L110E , PNP_SV45X45_L110E ,and PNP_SV135X135_L110E due to SPICE update. 3. Add MIMCAPS_20F_MML110E due to SPICE update. 4. Modified the TOP Metal resistor RS value 8K from 39E-3 ohm to 35E-3 ohm due to G-02-EDR update. 5. Modified the TOP Metal resistor RS value 20K from 20E-3 ohm to 15E-3 ohm due to G-02-EDR update. 6. Modified the METAL1 resistor Rs value from 196E-3 ohm to 170E-3 ohm due to G-02-EDR update. 7. Modified METAL2-METAL6 (1X Metal thickness) resistor Rs value from 147E-3 ohm to 130E-3 ohm due to G-02-EDR update.
0.1_p1_07	12/06/2010	Johnson Chou	1. Add BPW devices due to SPICE update 2. Modify the Boolean operation for 1.2V NMOS for SP option due to G-06 modified. 3. Modify the connection of inductor
0.1_p1_06	10/25/2010	Johnson Chou	1. ERC enhancement 2. Add PCAP_12, PCAP_33 3. Add CCI mapping file for StartRCXT 4. Add device stacking function
0.1_P1_05	09/06/2010	Johnson Chou	Update LVS deck due to DSM update 1. Add HS_SP_LL swithc for Mask-tooling update 2. Add device for SPICE update . RNHR1000_RF . MIMCAPS_1P5F_PSUB_RFKF . MIMCAPS_1P5F_NWEL_RFKF . MIMCAPS_1F_PSUB_RFKF . MIMCAPS_1F_NWEL_RFKF . LNW_SY40KCT_RFVIL . LNW_SY40K_RFVIL . LNW_CR40K_RFVIL . MOMCAPS_SY_MML110AE . MOMCAPS_AS_MML110AE . MOMCAPS_STRIPE_MML110AE . MOMCAPS_SYMESH_MML110AE . MOMCAPS_ASMESH_MML110AE . MOMCAPS_STRIPEMESH_MML111AE

			. MOMCAPS_ARRAY_VP5_RFVCL 3. Add MC switch for DSM added 4. Set LOD search distance to 30um. 5. Modify device relative parameter due to DSM (SPICE) change . RNPPO_RF . RNNPO_RF . MIMCAPS_15F_MML110E . MIMCAPS_10F_MML110E . RNNPO_L110E . RNPPO_L110E . RNND_L110E . RNPD_L110E . RNHR1000_L110E . RNWELL_MML110E 6. Change device name due to DSM (SPICE) change . L_SY40KNWCT_RFVIL --> LNW_SY40KCT_RFVIL . L_SY40KNW_RFVIL --> LNW_SY40K_RFVIL . L_CR40KNW_RFVIL --> LNW_CR40K_RFVIL
0.1_P1_04	03/29/2010	Momo Chang	1.Follow G-05 to add RF devices . 1.2V Pwell RF MOS (N_12_HSRF) . 1.2V Nwell RF MOS (P_12_HSRF) . 3.3V Pwell RF MOS (N_33_RF) . 3.3V Nwell RF MOS (P_33_RF) . 4 terminals RF N+ Poly sheet resistor (RNNPO_RF) . 4 terminals RF P+ Poly sheet resistor (RNPPO_RF) . 1.2V MIS varactor (VARMIS_12_RF) . 3.3V MIS varactor (VARMIS_33_RF) . P+/Nwel Junction Varactor (VARDIOP_RF) . (MOMCAPS_ARRAY_VP3_RF) . (MOMCAPS_ARRAY_VP4_RF) . RF's Planar MIM Capacitor (MIMCAPS_10F_RF) . RF's Planar MIM Capacitor (MIMCAPS_15F_RF) . RF's Planar MIM Capacitor (MIMCAPS_20F_RF) . Planar RF Inductor on PSUB (L_CR40K_RFVIL) . Planar RF Inductor on NWEL (L_CR40KNW_RFVIL) . Planar Symmetric RF Inductor on PSUB (L_SY40K_RFVIL) . Planar Symmetric RF Inductor on NWEL (L_SY40KNW_RFVIL) . Planar Symmetric center-tapped RF Inductor on PSUB (L_SY40KCT_RFVIL) . Planar Symmetric center-tapped RF Inductor on NWEL (L_SY40KNWCT_RFVIL) . Stacked RF Inductor on PSUB (L_SQSK_RFVIL) . Stacked RF Inductor on NWEL (L_SQSKNW_RFVIL) . Transformer without center taps (L_SQ_TRANS_RFVIL) . Transformer with center taps on primary coil (L_SQCTIN_TRANS_RFVIL) . Transformer with center taps on secondary coil (L_SQCTOUT_TRANS_RFVIL) . Octagonal Transformer with center taps on secondary coil (L_OCCTOUT_TRANS_RFVIL) . RF-PAD (PAD_RF) 2.Remove the structure of NMOS devices in Triple well.(please refer to NOTES 14) 3.Add NOTES 15 for RF resistors. 4.Remove LIB option setup
0.1_P1_03	03/12/2010	Momo Chang	1.Follow G-06 to modify NMK_* layer for 1.2V/3.3V NMOS/PMOS,BJT

			2.Add PVCМК layer for PVC cell device specification 3.Follow G-02 to modify parameter of METAL_RES_1X,METAL_RES_2X, 4.Modify derived layer:ME _x _I 5.Add connection for NCAP
0.1_P1_02	02/22/2010	Momo Chang	1.Follow G-05 to add NCAP devices (NCAP_12/NCAP_33) 2.Follow G-05 to add IG MOS: (N_12_HSL110E_IG/P_12_HSL110E_IG/N_12_SPL110E_IG/P_12_SP L110E_IG/N_12_LLL110E_IG/P_12_LLL110E_IG) 3.Follow G-05 & G-06 to add BJT devices (PNP_V45X45_L110E/PNP_V90X90_L110E/PNP_V135X135_L110E) 4.Follow G-06 to modify derived layer:NMK_PNP,NMK_N_NVT_12,NMK_N_NVT_33, 5.Add derived layer:ME _x _I,NSD_I,PSD_I 6.Modify MIMCAP devices's assignment in different Metal Option.
0.1_P1_01	01/15/2010	Momo Chang	New Create

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General Description

This version of command file has passed the quality assurance checking, and the design verification has been made. It follows the information below.

LVS Deck Package Deliverables

There are 12 files in this deck package deliverables, as description below:

- LVS deck release notes
[Release_Notes_](#)
[LOGIC_MIXED_MODE11-1P8M-MMC_AL_L110AE-CALIBRE-LVS-1.2_P4.pdf](#)
- LVS application notes.
[Application_Notes_](#)
[LOGIC_MIXED_MODE11-1P8M-MMC_AL_L110AE-CALIBRE-LVS-1.2_P4.pdf](#)
- LVS QA report.
[QA_Report_](#)
[LOGIC_MIXED_MODE11-1P8M-MMC_AL_L110AE-CALIBRE-LVS-1.2_P4.pdf](#)
- CCI and QCI mapping file, there are 2 files as shown below
[G-DF-LOGIC_MIXED_MODE11-1P6M-MMC-AL_L110AE-CALIBRE-LVS_MAP_FOR_CCI_1.2_P4.txt](#)
[G-DF-LOGIC_MIXED_MODE11-1P6M1T-MMC56-AL_L110AE-CALIBRE-LVS_MAP_FOR_QCI_1.2_P4.txt](#)
- Main LVS Deck files, there are 1 files as shown below
[G-DF- LOGIC_MIXED_MODE11-1P8M-MMC_AL_L110AE-CALIBRE-LVS-1.2_P4.txt](#)
- LPE command files, there are 6 files as shown below
CCI :
[CCI_star_cmd](#)
[Query.txt](#)
QCI:
[QCI_qrc_cmd](#)
[QCI_setup_readme.txt](#)
[query_QCI.txt](#)
[techgen_scr](#)